Constrained Bandit Learning with Switching Costs for Wireless Networks

Juaren Steiger† Bin Li‡ Bo Ji†† Ning Lu†

†Department of Electrical and Computer Engineering, Queen's University, Kingston, Ontario, Canada
‡School of Electrical Engineering and Computer Science, Pennsylvania State University, State College, Pennsylvania, USA
††Department of Computer Science, Virginia Tech, Blacksburg, Virginia, USA
Email: {juaren.steiger, ning.lu}@queensu.ca, binli@psu.edu, boji@vt.edu

IEEE INFOCOM 2023 - IEEE Conference on Computer Communications | 979-8-3503-3414-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/INFOCOM53939.2023.10228986

***Abstract*—Bandits with arm selection constraints and bandits with switching costs have both gained recent attention in wireless networking research. Pessimistic-optimistic algorithms, which combine bandit learning with virtual queues to track the constraints, are commonly employed in the former. Block-based algorithms, where switching is disallowed within a block, are commonly employed in the latter. While efficient algorithms have been developed for both problems, it remains challenging to guarantee low regret and constraint violation in a bandit problem that includes both arm selection constraints and switching costs due to the tight coupling between the two. Here, switching may be necessary to decrease the constraint violation but comes at the cost of increased switching regret. In this paper, we tackle the constrained bandits with switching costs problem, for which we design a block-based pessimistic-optimistic algorithm. We identify three timely wireless networking applications for this framework in edge computing, mobile crowdsensing, and wireless network selection. We also prove that our algorithm achieves sublinear regret and vanishing constraint violation and corroborate these results with synthetic simulations and extensive trace-based simulations in the wireless network selection setting.**

I. Introduction

The *stochastic multi-armed bandit* problem and its variants have recently attracted much attention as frameworks for modelling problems in wireless networking and other computer communication related fields. For example, Li et al. have compiled a survey of bandit algorithms for spectrum scheduling in wireless networks [1], Zaheer et al. discuss bandit learning approaches in their survey of *cognitive Internet of things* [2], as do Caso et al. in their survey of radio access technology selection [3]. In the basic stochastic bandit problem, the learning agent (or *player*) must choose from a set of actions (called *"arms"*) that each yields *reward* drawn independently and identically distributed (i.i.d.) over time from distributions with unknown statistics. This problem presents the simplest possible instance of the *exploration vs. exploitation dilemma* in reinforcement learning, in which the player would like to *exploit* the currently most promising arms, while *exploring* underplayed arms which may potentially be more rewarding. In this problem, we are interested in minimizing *regret*, i.e. the cumulative expected difference in reward between the player's policy and the optimal policy when the statistics are known. Two extensions of the classical multi-armed bandit problem have recently emerged in the wireless networking literature, namely *arm selection constraints* and *arm switching costs*.

In their discussion on future directions for bandit algorithms applied to wireless spectrum scheduling, Li et al. mention "budget constraints" that consider the overhead of playing different arms. On the other hand, arms may represent users or other entities that wish to be treated *fairly* in some problems, motivating constraints that each arm should be played sufficiently often. These various *arm selection constraints* are the first of the two extensions relevant to wireless networking. State-of-the-art work in this direction considers strong constraints that hold cumulatively in each timeslot, rather than as long-term averages [4]–[7]. Liu et al. [6] call these *"anytime cumulative constraints"*. Then, in addition to minimizing regret in such constrained bandit problems, we are also interested in minimizing the *cumulative constraint violation*.

A commonly employed algorithm design in this type of problem is to maintain virtual queues to track the constraint violation and combine the *UCB algorithm* [8] for bandit learning with the commonly used *MaxWeight* algorithm [9] to stabilize the virtual queues. Liu et al. [6] call this style of algorithm a *"pessimistic-optimistic"* algorithm because the UCB gives an *optimistic* overestimate of the mean reward of each arm and each virtual queue gives a *pessimistic* overestimate of the violation of each constraint. State-of-the-art pessimistic-optimistic algorithms [6] achieve $\widetilde{O}(\sqrt{T})$ regret up to some time horizon T and $o(1)$ constraint violation. [1] In other words, the regret matches the instance-independent bound for UCB [10] and the constraint violation vanishes.

In many applications, arms represent resources that incur significant computation or communication overhead to switch out for other arms. The second extension to the multi-armed bandit problem that has attracted recent attention in the wireless networking literature is then to consider *arm switching costs*. In the bandits with switching costs problem, we are interested in maximizing reward, while minimizing the number

[1]For positive functions f and g, the notation $f(x) = o(g(x))$ means $\lim_{x\to\infty} \frac{f(x)}{g(x)} = 0$, $f(x) = O(g(x))$ means $\limsup_{x\to\infty} \frac{f(x)}{g(x)} < \infty$, and $f(x) = \Omega(g(x))$ means $\liminf_{x\to\infty} \frac{f(x)}{g(x)} > 0$. The notation $f(x) = \widetilde{O}(g(x))$ and $f(x) = \widetilde{\Omega}(g(x))$ ignore logarithmic factors in g.

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.

of arm switches. Thus, regret in this problem is comprised of the usual difference in reward (*reward regret*) as well as a cumulative switching (*switching regret*) term. Deka et al. [11] prove a $\widetilde{\Omega}(T^{2/3})$ lower bound on the minimax regret of bandits with switching costs.

A typical algorithm design approach to address this switching problem is to divide time into blocks of increasing length. Then the player makes a decision at the beginning of each block and sticks with it for the remainder of the block. So, the player only suffers switching cost for one timeslot per block, and these timeslots become increasingly far apart. There has existed an asymptotically optimal ($O(\log T)$ regret) block-based UCB algorithm for stochastic bandits with switching costs since the introduction of the problem [12]. The state-of-the-art block-based algorithms achieve $O(T^{2/3})$ regret in the more general adversarial bandit problem (matching the minimax lower bound) and $O(T^{1/3})$ regret in the stochastic bandit problem [13], [14]. In addition to block-based algorithms, other types of algorithms have been developed for this problem, such as the Gittins index policy designed by Asawa et al. [15] for the infinite (discounted) time horizon case.

Simultaneously minimizing the reward regret, switching regret, and constraint violation proves to be very challenging in the combined *"constrained bandits with switching costs"* problem. This is because exploration increases both the switching regret and the constraint violation. Also, switching is necessary to decrease the reward regret (via exploration) and to decrease the constraint violation. So far, little progress has been made toward guaranteeing sublinear reward regret, switching regret, and constraint violation simultaneously. Recent work in this direction [16], [17] instead focuses on bounding the time-average switching cost by a constant, which implies linear switching regret.

In this paper, we consider this difficult constrained bandits with switching costs problem motivated by applications in wireless networking (see Section III) and develop an algorithm that achieves sublinear regret and vanishing constraint violation. In the following subsections, we present the recent work in wireless networking research related to bandits with arm selection constraints and bandits with switching costs and then discuss our contributions in detail.

A. Related Work

The study of bandits with arm selection constraints started with the work of Li et al. [18] who considered *fairness constraints* where each arm should be played a minimum fraction of the time, motivated by applications in network resource allocation, online advertising, and crowdsourcing. They consider long-term constraints and use the Lyapunov Drift Theorem [19] to prove long-term queue stability, which implies $o(t)$ constraint violation of the anytime cumulative version of these constraints. Since this seminal work [18], pessimistic-optimistic algorithms have been widely applied, for example, in the areas of IoT [20], caching networks [21], mobile crowdsensing [22], and federated learning [23], [24]. These applications include both fairness constraints, and similar per-arm cost constraints. Recently, Liu et al. formulated a more general bandits with arm selection constraints framework in which these various fairness and cost constraints can be encoded [6]. Moreover, the number of constraints is completely decoupled from the number of arms in their model, which they address in their pessimistic-optimistic algorithm by weighting the virtual queue for each constraint by the contribution of each arm to the constraint. They also add a "tightness" parameter to the virtual queues to control the instantaneous constraint violation and show that it eventually vanishes (an improvement from $o(t)$ to $o(1)$). Zhou et al. [7] consider (kernelized) bandits with constraints, with [4] considering an application to beam alignment in mmWave communications. $O(\sqrt{T})$ regret and constraint violation is achieved in this setting.

Bandits with switching costs have also attracted much attention in the wireless networking literature. For example, Zhou et al. [25] consider bandit learning to decide which small base station a user will connect to subject to base station handover (switching) costs. Rejiba et al. [26] consider bandit learning for fog node selection where switching costs model the time to start up a new container and Zhu et al. [27] consider task offloading to fog helper nodes where switching costs model energy consumption and latency due to the initialization of a communication link. Le et al. [28] consider wireless channel monitoring where switching costs are incurred when sniffers are reassigned to different channels. Li et al. [29] consider participant selection in crowdsensing with switching costs. Alipour-Fanid et al. [30] consider multi-user scheduling with channel switching costs and Appavoo et al. [31] consider wireless network selection for a mobile user with network switching costs. Guha and Munagala motivate their study of bandits with metric switching costs with an application to wireless sensor networks [32].

Some recent work has been done in constrained bandit learning problems with switching costs [16], [17]. For example, Krishnasamy et al. [16] consider the problem of switching base stations on and off in a cellular network to save energy, while guaranteeing the required service to users. They are able to guarantee long-term constraints and small constant time-average cumulative switching. But, to our knowledge, no work has been able to achieve sublinear (switching and reward) regret and vanishing constraint violation in such a problem.

B. Our Contributions

In this paper, we consider bandits with switching costs subject to arm selection constraints in the form of *anytime cumulative constraints*. We then identify three timely applications in the domains of edge computing, mobile crowdsensing, and wireless network selection (see Section III). To solve this problem, we develop a block-based pessimistic-optimistic algorithm called *Pessimistic-Optimistic-Slow-Switching* (POSS) and analyze its regret and constraint violation performance. Finally, we compare our algorithm against others from the literature with both synthetic and trace-based simulations in the wireless network selection setting.

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.

While both the block-based and pessimistic-optimistic algorithm design approaches have been well studied, combining the two is nontrivial. A block-based algorithm needs to be analyzed on the timescale of blocks, since the decision-making process happens on this timescale. Thus, we need to design the virtual queues to be updated from block-to-block instead of timeslot-to-timeslot. However, this is challenging as the length of blocks increases over time. Then we will not be able to bound the block-to-block drift of the virtual queues by a constant, as is usually done in analyses following Hajek [33] (see e.g. [5], [6], [34]). Therefore, we need to develop novel drift lemmas in order to bound the constraint violation. Also, designing the block size schedule is a challenge for our problem. For example, we cannot use the exponentially increasing block sizes from Agrawal et al. [12], as done in other work (e.g. [29], [35]) since this relies on an instance-dependent analysis that bounds the number of plays of a suboptimal arm. However, when considering the arm selection constraints, such an approach is intractable since every arm may continually need to be played to satisfy the constraints. Finally, the parameters for the pessimistic and optimistic components of the algorithm need to be carefully chosen so that the algorithm performs well compared to non-block-based pessimistic-optimistic algorithms. Through careful design of our algorithm, it is provably able to achieve $\widetilde{O}(T^{3/4})$ regret and vanishing constraint violation, i.e. $o(1)$. These results are corroborated by synthetic and trace-based simulations.

The rest of this paper is organized as follows. In Section II, we introduce the constrained bandits with switching costs problem. In Section III, we identify three timely wireless networking applications of this problem. In Section IV, we present our algorithm and in Section V, we analyze its regret and constraint violation. In Section VI, we discuss our synthetic and trace-based simulation results. Finally, in Section VII, we conclude the paper and discuss future research directions.

II. Problem Setting

We consider a stochastic multi-armed bandit problem with N arms indexed by $i = 1, 2, \ldots, N$, each associated with an i.i.d. reward process $X_i(t) \in [0, 1]$ with unknown a priori mean μ_i. We also consider K different *anytime cumulative constraints* indexed by $k = 1, 2, \ldots, K$, each characterized by N coefficients $w_{i,k} \in [-1, 1]$ for $i \in [N]$. Here, $w_{i,k}$ represents a penalty associated with constraint k of playing arm i in timeslot t and we require that the cumulative penalty be nonpositive for each constraint k in each timeslot t. As an example of anytime cumulative constraints, the fairness constraints introduced by Li et al. [18] can be encoded by considering $K = N$ constraints characterized by the coefficients $w_{i,k} = r_k - \mathbb{1}\{i = k\}$ where r_i is the minimum selection fraction associated with arm i.

The *constrained bandits with switching costs* problem proceeds as follows. In each timeslot $t = 1, 2, \ldots$, the player plays an arm $i(t) \in [N]$, receives the reward $X_{i(t)}(t)$ as feedback and suffers a penalty $w_{i(t),k}$ for each constraint k. The player's objective is then to play arms that maximize the cumulative expected reward $\mathbf{E}[X_{i(1)}(1)] + \mathbf{E}[X_{i(2)}(2)] + \cdots$, while ensuring that $\sum_{\tau=1}^{t} \mathbf{E}[w_{i(\tau),k}] \leq 0$ in each timeslot t for each constraint k.

In addition to anytime cumulative constraints, we also consider a cost $C > 0$ of switching arms. In other words, the player suffers a switching cost C in timeslot t in the event that $i(t) \neq i(t-1)$. We indicate this event by $H(t) \triangleq \mathbb{1}\{i(t) \neq i(t-1)\}$ for $t > 1$, with $H(1) = 0$. Then for a *time horizon* T, the *constrained bandits with switching costs* problem is given by

$$\begin{aligned} \underset{\{i(t)\in[N]\}_{t=1}^{T}}{\text{maximize}} \; & \sum_{t=1}^{T} \mathbf{E}\big[X_{i(t)}(t) - C\, H(t)\big] \\ \text{subject to} \; & \sum_{\tau=1}^{t} \mathbf{E}[w_{i(\tau),k}] \leq 0 \quad \forall\, k \in [K],\ t = 1, 2, \ldots \end{aligned} \tag{1}$$

We assume that Slater's condition holds for the above problem. That is, there exists a positive constant $\gamma \in (0, 1]$ (the *Slater's constant*) and a stochastic N-vector $\boldsymbol{\sigma}^{\gamma}$ such that

$$\sum_{i\in[N]} \sigma_i^{\gamma} w_{i,k} \leq -\gamma \quad \forall\, k \in [K]. \tag{2}$$

In the proceeding subsection, we show how a Markovian policy can solve the problem (1) optimally, assuming the knowledge of the mean rewards $\boldsymbol{\mu}$. However, in the online learning problem with unknown $\boldsymbol{\mu}$, it is not reasonable to expect that a learning algorithm can satisfy the anytime cumulative constraints. Instead, we evaluate a policy's adherence to the constraints by considering the *constraint violation* in timeslot t for each constraint k given by

$$\mathcal{V}_k(t) \triangleq \left[\sum_{\tau=1}^{t} \mathbf{E}[w_{i(\tau),k}]\right]^{+} \tag{3}$$

where the notation $[\,\cdot\,]^{+} \triangleq \max\{\,\cdot\,, 0\}$.

A. The Optimal Markovian Policy and Regret

Consider a randomized policy π that plays $i^{\pi}(t) \in [N]$ according to a Markov chain with $N \times N$ stochastic matrix $\mathbf{P}^{\pi}$ and stationary distribution $\boldsymbol{\sigma}^{\pi}$. This policy π plays $i^{\pi}(1) = i$ with probability σ_i^{π} and $i^{\pi}(t) = i$ with probability $P^{\pi}_{i(t-1),i}$ in timeslots $t > 1$. Then $i^{\pi}(t)$ is a Markov chain in steady state. We will call such a policy π a *Markovian policy*.

A crucial observation in constructing Markovian policies [16] is that if we are given a stochastic N-vector $\boldsymbol{\sigma}$, we can construct a stochastic matrix $\mathbf{P}^{(\boldsymbol{\sigma},s)}$ parameterized by $\boldsymbol{\sigma}$ and a constant $s \in [0, 1]$ such that $\boldsymbol{\sigma}$ is its stationary distribution by setting

$$P_{i,j}^{(\boldsymbol{\sigma},s)} \triangleq \begin{cases} (1-s) + s\,\sigma_i & \text{for } i = j \\ s\,\sigma_j & \text{for } i \neq j \end{cases} \tag{4}$$

for all $(i, j) \in [N]^2$. It's easy to check that $\mathbf{P}^{(\boldsymbol{\sigma},s)}$ is indeed a stochastic matrix and that $\boldsymbol{\sigma}\mathbf{P}^{(\boldsymbol{\sigma},s)} = \boldsymbol{\sigma}$. The parameter s here tunes the probability of switching arms. At one end,

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.

$s = 1$ switches freely and at the other, $s = 0$ guarantees no switching.

Then we can construct the optimal Markovian policy $i^*(t)$ by first finding the stochastic vector $\boldsymbol{\sigma}^*$ that solves the linear program

$$\begin{aligned} \underset{\text{stochastic } N\text{-vector } \boldsymbol{\sigma}}{\text{maximize}} \quad & \sum_{i \in [N]} \sigma_i \mu_i \\ \text{subject to} \quad & \sum_{i \in [N]} \sigma_i w_{i,k} \leq 0 \quad \forall\, k \in [K] \end{aligned} \tag{5}$$

and then setting $\mathbf{P}^* = \mathbf{P}^{(\boldsymbol{\sigma}^*, 0)}$. To see why $\mathbf{P}^*$ is optimal, note that the Markovian policy characterized by $\boldsymbol{\sigma}^*$ and $\mathbf{P}^{(\boldsymbol{\sigma}^*, 1)}$ is known to be optimal in the constrained bandit problem *without* switching costs [6], [18], and substituting $s = 1$ with any $s \in [0, 1]$ achieves the same expected reward. Moreover, setting $s = 0$ guarantees that no switching cost is incurred in the constrained bandit problem *with* switching costs, so $\mathbf{P}^*$ is optimal in this setting.

Then, using $i^*(t)$ as a benchmark for our algorithm, we aim to minimize the *cumulative regret* given by

$$R(T) \triangleq \underbrace{\sum_{t=1}^{T} \mathbf{E}\big[X_{i^*(t)}(t) - X_{i(t)}(t)\big]}_{\text{reward regret } R^{\texttt{reward}}(T)} + \underbrace{\sum_{t=1}^{T} \mathbf{E}[C\, H(t)]}_{\text{switching regret } R^{\texttt{switch}}(T)} \quad . \tag{6}$$

B. Problem Extensions

The problem we present in this section can be extended in a few different ways. We choose to omit these extensions from the presentation, so as not to overcomplicate the exposition. However, these extensions can be accommodated by slightly adjusting our solution and do not significantly impact the analysis of our algorithm.

Combinatorial Semi-Bandit Feedback. Many real-world applications have decisions and feedback that are combinatorial in nature [5], [18]. In this feedback regime, the player plays a collection of $M < N$ arms, called a *super-arm* $\mathcal{S}(t)$, in each timeslot t instead of a single arm and receives the reward from each arm in the super-arm individually, i.e. $\{X_i(t)\}_{i \in \mathcal{S}(t)}$. The switching cost in this extension may be defined, for example, by scaling C by the cardinality of the symmetric difference of $\mathcal{S}(t)$ and $\mathcal{S}(t-1)$. Our algorithm can be adjusted to accommodate this extension by simply choosing the super-arm with the greatest sum of pessimistic-optimistic indices, as done in [5], [18], which shouldn't affect the asymptotic bounds achieved by our algorithm.

Switching Delay that Impacts Reward and Constraints. Another real-world applicable extension is the case where the player receives no reward and suffers constraint penalty for a fixed number of timeslots δ after switching. Since we employ a block-based algorithm, where the block-size is always increasing, this period δ becomes negligible compared to the size of each block and does not affect our asymptotic results. This extension can be accommodated by multiplying the size of each block by $\delta + 1$.

Random Constraint Coefficients. The problem is easily extended to the case where instead of constant constraint coefficients $\mathbf{w}$, we have random variables $\mathbf{W}(t)$ with a priori unknown statistics that are revealed to the player at the beginning of each round (as in the problem from Liu et al. [6]). Then we need only replace $w_{i,k}$ with $W_{i,k}(t)$ in our algorithm.

However, a nontrivial extension is the case where the random variables $W_{i(t),k}(t)$ for each k are revealed to the player *at the end* of each round as bandit feedback along with the reward. This is because the virtual queues in the pessimistic-optimistic algorithm are interdependent with the estimates for the constraint coefficients, which makes the analysis of the *expected* constraint violation extremely difficult (see Appendix D in [7] for an in depth discussion).

III. Wireless Networking Applications

In this section, we present three timely wireless networking applications to motivate the study of constrained bandits with switching costs. The third application will be the focus of our trace-based simulations.

A. Machine Learning Model Deployment on an Edge Server

Following the application setting presented by Shi et al. [36], consider an E-commerce recommendation system in which an edge server with limited storage downloads and deploys product recommendation models from a remote cloud, which it uses to recommend products to incoming customers. The edge server can store at most M of the N available models at a time (see Section II-B) and each model has a priori unknown predictive capability. Here, a bandit arm is a model $i \in [N]$, and reward $X_i(t) \in \{0, 1\}$ is Bernoulli and indicates when a user clicks on a recommended product. However, switching models incurs communication and model deployment overhead, which is modeled by the switching cost C. Furthermore, each model i has an energy consumption rate e_i and computation delay d_i and we allow a maximum average energy consumption $e_{\max}$ and computation delay $d_{\max}$. Then these $K = 2$ constraints are characterized by $w_{i,1} = e_i - e_{\max}$ and $w_{i,2} = d_i - d_{\max}$ for each model i.

B. Participant Selection in Mobile Crowdsensing

Li et al. [29] identify the problem of participant selection for tasks in a mobile crowdsensing platform. Here, the bandit arms are task-to-user assignments, which are combinatorial in nature (see Section II-B), i.e. $N =$ # of users $\times$ # of tasks, each arm $i \in [N]$ is a user-task pair, and a super-arm is a matching between users and tasks. Reward $X_i(t) \in [0, 1]$ for arm i is the completed portion of the task by the user in timeslot t, which depends on the random and unknown a priori mobility and capability of the user. When a user is assigned to a task, they enter into a contract with the crowdsensing platform. Then there is a switching cost C associated with assigning a user a different task in the form of monetary compensation for breaking the contract. Furthermore, we impose fairness constraints for the participants (i.e. $K =$ # of users) because

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.

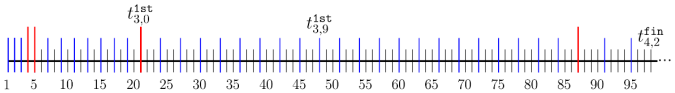


Fig. 1. A demonstration of blocks and frames for $N = 3$ arms. A tall red line represents a new frame and a medium blue line represents a new block.

participants that are never selected will lose interest in the platform or generate bad publicity.

C. Wireless Network Switching for a Mobile User

Consider a mobile user that may connect to one of N possible wireless networks, e.g. the 5G cellular network or different Wi-Fi access points [31], [37] where each network $i \in [N]$ can transmit a packet with probability μ_i in each timeslot t. Then $X_i(t) \sim \text{Bernoulli}(\mu_i)$ indicates the potential service from network i in timeslot t and acts as the reward. Note that in reality, the service for each network depends on the number of connected users, but from a single user's perspective, this can be considered i.i.d. We assume that when the user's queue backlog of packets is empty, they generate a dummy packet to probe the network. So, the service $X_i(t)$ may always be observed. In each timeslot t, the user can stay connected to network $i(t) = i(t-1)$ or switch to a network $i(t) \neq i(t-1)$ and suffer a switching cost C that models the resulting latency and service interruption. Note that in reality, switching networks is not instantaneous, so this application is more realistic under the switching delay extension we discuss in Section II-B. Additionally, different networks are known to affect the battery usage of the mobile device differently. We assume that using network i is known to incur an average battery consumption of $\beta_i \in [0,1]$ per timeslot and the user allows a maximum battery consumption rate of $\beta_{\max}$. Typically 5G can guarantee greater service than Wi-Fi but at the cost of greater battery usage [38], [39]. Then we have a battery consumption constraint characterized by $w_{i,k} = \beta_i - \beta_{\max}$ for each network i. We could also impose a monetary constraint since using mobile data incurs a monetary cost in a pay-as-you-go style plan, whereas Wi-Fi is typically free to use.

IV. Algorithm Design

We design a block-based pessimistic-optimistic algorithm, called *Pessimistic-Optimistic-Slow-Switching (POSS)*, to control both the switching and constraint violation simultaneously. Here, a decision of which arm to play is made greedily at the beginning of each block based on each arm's *pessimistic-optimistic index*. This index consists of an optimistic term, i.e. the UCB estimate of each arm's mean reward, and a pessimistic term, i.e. the virtual queues tracking the constraints. To guarantee sublinear cumulative switching, the block size must increase over time. We accomplish this by grouping blocks into *frames* such that the block size increases linearly from frame-to-frame and the number of blocks in each frame increasing polynomially. We first explain the mechanics of blocks and frames in detail, and then present the components necessary for the pessimistic-optimistic index.

A. Blocks and Frames

We group timeslots into frames indexed by $f = 0, 1, 2, \ldots$ and divide each frame f into blocks indexed by $b = 0, 1, 2, \ldots, b_f$, with b_f denoting the last block in frame f. We denote the first timeslot in block b of frame f by $t^{\mathtt{1st}}_{f,b}$ and the final timeslot in block b of frame f by $t^{\mathtt{fin}}_{f,b}$. Switching can only be done in the first timeslot in a block, meaning that $i(t^{\mathtt{1st}}_{f,b}) = i(t^{\mathtt{1st}}_{f,b} + 1) = \cdots = i(t^{\mathtt{fin}}_{f,b})$ for each block b in each frame f.

Frame $f = 0$ contains N blocks, each containing one timeslot. In this frame, we use each block to explore each arm once, i.e. we choose arm i in block $b = i - 1$. For the proceeding frames $f > 0$, each block b in frame f contains f timeslots and each frame f contains $\left\lceil \left(f^4 - (f-1)^4\right)/f \right\rceil$ blocks. Figure 1 gives a demonstration of the mechanics of blocks and frames.

Remark 1. $t^{\mathtt{fin}}_{f,b_f} = N + \sum_{\ell=1}^{f} \left\lceil \left(\ell^4 - (\ell-1)^4\right)/\ell \right\rceil \ell$, i.e. the final timeslot of each frame $f > 0$ is upper bounded by

$$t^{\mathtt{fin}}_{f,b_f} \leq N + \underbrace{\sum_{\ell=1}^{f} \left(\ell^4 - (\ell-1)^4\right)}_{=f^4} + \underbrace{\sum_{\ell=1}^{f} \ell}_{=f(f+1)/2} \leq 3Nf^4 \tag{7}$$

and lower bounded by

$$t^{\mathtt{fin}}_{f,b_f} \geq \sum_{\ell=1}^{f} \left(\ell^4 - (\ell-1)^4\right) = f^4. \tag{8}$$

These bounds are useful for changing timescales between frames and timeslots.

B. Pessimistic-Optimistic Index

To construct the optimistic component of our algorithm, we will need to maintain an empirical mean $\hat{\mu}_i(t)$ to estimate each μ_i. Note that after frame $f = 0$, we will have observed at least one realization of the reward from each arm i. Therefore, each $\hat{\mu}_i(t)$ is well defined for timeslots $t > t^{\mathtt{fin}}_{0,b_0}$. Then the optimistic component is given by the UCB estimate of each μ_i defined as

$$U_i(t) \triangleq \min\left\{\hat{\mu}_i(t) + \sqrt{\frac{3\log t}{2n_i(t)}}, 1\right\} \tag{9}$$

where $n_i(t) \triangleq \sum_{\tau=1}^{t-1} \mathbb{1}\{i(\tau) = i\}$ is the number of times arm i has been played up to timeslot t.

Next, we introduce the pessimistic components of the algorithm. These are the *block-based virtual queues* $\{Z_k(t^{\mathtt{1st}}_{f,b})\}_{k\in[K]}$ that track the constraints from block-to-block and are defined only for the subsequence of timeslots $\{t^{\mathtt{1st}}_{f,b}\}_{b=0,1,\ldots,b_f,\, f=1,2,\ldots}$. After the frame $f = 0$ reserved for initial exploration, we initialize each block-based virtual queue at the beginning of frame $f = 1$ to $Z_k(t^{\mathtt{1st}}_{1,0}) = \left[\sum_{\tau=1}^{N} w_{i(\tau),k}\right]^+$. In timeslot $t^{\mathtt{fin}}_{f,b} + 1$ (which is equal to

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.

timeslot $t^{\mathtt{1st}}_{f,b+1}$ if $b < b_f$ or $t^{\mathtt{1st}}_{f+1,0}$ if $b = b_f$), the block-based virtual queue tracking constraint k is updated as

$$Z_k(t^{\mathtt{fin}}_{f,b}+1) = \left[Z_k(t^{\mathtt{1st}}_{f,b}) + \sum_{\tau=t^{\mathtt{1st}}_{f,b}}^{t^{\mathtt{fin}}_{f,b}} w_{i(\tau),k} + \varepsilon_f \right]^+. \quad (10)$$

$\{\varepsilon_f\}_{f=1,2,\ldots}$ is a real-valued sequence that controls the constraint violation given by

$$\varepsilon_f \triangleq \frac{f^3 \log^{\frac{1}{3}}(f+1) - (f-1)^3 \log^{\frac{1}{3}} f}{\lceil (f^4 - (f-1)^4)/f \rceil}. \quad (11)$$

It is important to note that each $\varepsilon_f \in (0, f]$ and that the sequence $\{\varepsilon_f/f\}_{f=1,2,\ldots}$ is decreasing.

Remark 2. We would like to give some intuition for the sequence $\{\varepsilon_f\}$. Expanding the update equation (10) gives

$$\begin{aligned} \mathbf{E}[Z_k(t^{\mathtt{1st}}_{f,b})] &\geq \sum_{\tau=1}^{t^{\mathtt{1st}}_{f,b}-1} \mathbf{E}\big[w_{i(\tau),k}\big] + \sum_{\ell=1}^{f-1}\sum_{b=0}^{b_\ell} \varepsilon_\ell + b\,\varepsilon_f \\ &\overset{(a)}{\geq} \sum_{\tau=1}^{t^{\mathtt{1st}}_{f,b}-1} \mathbf{E}\big[w_{i(\tau),k}\big] + (f-1)^3 \log^{\frac{1}{3}} f \end{aligned} \quad (12)$$

where (a) is because $b\,\varepsilon_f \geq 0$ and

$$\begin{aligned} \sum_{\ell=1}^{f-1}\sum_{b=0}^{b_\ell} \varepsilon_\ell &= \sum_{\ell=1}^{f-1} \Big(\ell^3 \log^{\frac{1}{3}}(\ell+1) - (\ell-1)^3 \log^{\frac{1}{3}} \ell \Big) \\ &= (f-1)^3 \log^{\frac{1}{3}} f. \end{aligned} \quad (13)$$

Then for any timeslot t in block b of frame f, we have

$$\begin{aligned} \mathcal{V}_k(t) &\leq \left[\sum_{\tau=1}^{t^{\mathtt{1st}}_{f,b}-1} \mathbf{E}\big[w_{i(\tau),k}\big] + \sum_{\tau=t^{\mathtt{1st}}_{f,b}}^{t^{\mathtt{fin}}_{f,b}} \big[\mathbf{E}\big[w_{i(\tau),k}\big]\big]^+ \right]^+ \\ &\leq \Big[\mathbf{E}[Z_k(t^{\mathtt{1st}}_{f,b})] - (f-1)^3 \log^{\frac{1}{3}} f + f \Big]^+. \end{aligned} \quad (14)$$

Here, we can see the role of the sequence $\{\varepsilon_f\}$ in controlling the constraint violation.

Now, we have presented all the components necessary to define the pessimistic-optimistic index of arm i at the beginning of block b, which is given by

$$\mathtt{PO}_i(t^{\mathtt{1st}}_{f,b}) \triangleq f^3\, U_i(t^{\mathtt{1st}}_{f,b}) - f \sum_{k\in[K]} w_{i,k}\, Z_k(t^{\mathtt{1st}}_{f,b}). \quad (15)$$

Then, at the beginning of each block b in each frame f, the POSS algorithm plays arm $i(t^{\mathtt{1st}}_{f,b}) = i(f,b)$ where

$$i(f,b) \in \operatorname*{argmax}_{i\in[N]} \mathtt{PO}_i(t^{\mathtt{1st}}_{f,b}). \quad (16)$$

V. Theoretical Results

In this section, we present our theoretical results for the POSS algorithm. First, we show that POSS achieves vanishing constraint violation, i.e. $\mathcal{V}_k(t) = o(1)$ for each constraint k. Second, we show that POSS achieves $R^{\mathtt{reward}} = \widetilde{O}(T^{3/4})$ and $R^{\mathtt{switch}} = O(T^{3/4})$.

A. Vanishing Constraint Violation

Our approach to bound the constraint violation was previously used by the authors in [6] and [5] and originates in the work of Hajek [33]. This technique involves deriving drift conditions on the Euclidean norm of the virtual queues and using these conditions to bound its moment-generating function. The difference here from previous work is that the relevant drift bounds are time-dependent (frame-dependent) instead of constant and therefore we require a novel bound on the moment-generating function that is increasing with time instead of constant. We then rely on the sequence $\{\varepsilon_f\}$ to offset this bound and get vanishing constraint violation.

We will make extensive use of two Lyapunov functions in our analysis of the block-based virtual queues, namely the Euclidean norm $\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\|$ and the quadratic Lyapunov function $V(t^{\mathtt{1st}}_{f,b}) \triangleq \frac{1}{2}\sum_{k\in[K]} Z_k^2(t^{\mathtt{1st}}_{f,b})$. Let Δ denote the forward difference operator between blocks. For example, $\Delta\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\| = \|\mathbf{Z}(t^{\mathtt{fin}}_{f,b}+1)\| - \|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\|$.

Remark 3. We will frequently use the following bound on the drift of the quadratic Lyapunov function for frames $f \geq 1$:

$$\begin{aligned} &\Delta V(t^{\mathtt{1st}}_{f,b}) \\ &\leq \frac{1}{2}\sum_{k\in[K]} \left(Z_k(t^{\mathtt{1st}}_{f,b}) + \sum_{\tau=t^{\mathtt{1st}}_{f,b}}^{t^{\mathtt{fin}}_{f,b}} w_{i(\tau),k} + \varepsilon_f \right)^2 - V(t^{\mathtt{1st}}_{f,b}) \\ &= \sum_{k\in[K]} Z_k(t^{\mathtt{1st}}_{f,b}) \left(f w_{i(f,b),k} + \varepsilon_f \right) \\ &\quad + \frac{1}{2}\sum_{k\in[K]} \left(f w_{i(f,b),k} + \varepsilon_f \right)^2 \\ &\overset{(a)}{\leq} \sum_{k\in[K]} Z_k(t^{\mathtt{1st}}_{f,b}) \left(f w_{i(f,b),k} + \varepsilon_f \right) + 2Kf^2 \end{aligned} \quad (17)$$

where (a) is because each $w_{i(f,b),k} \in [-1,1]$ and $\varepsilon_f \in (0,f]$, and so $\left| f w_{i(f,b),k} + \varepsilon_f \right| \leq 2f$.

As noted by Eryilmaz et al. [40], the square root function can be bounded by its first-order Taylor approximation because it is concave and differentiable, i.e. $\sqrt{x_1} - \sqrt{x_2} \leq (x_1 - x_2)\frac{\partial\sqrt{x_2}}{\partial x_2} = \frac{x_1 - x_2}{2\sqrt{x_2}}$ for all $x_1, x_2 > 0$. Therefore, we have the following connection between the Euclidean norm and the quadratic Lyapunov functions:

$$\Delta\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\| = \Delta\sqrt{2\,V(t^{\mathtt{1st}}_{f,b})} \leq \frac{\Delta V(t^{\mathtt{1st}}_{f,b})}{\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\|}. \quad (18)$$

Using these facts, we now derive the drift conditions on the Euclidean norm of the virtual queues.

Lemma 1. *The following conditions hold for frames $f > 0$ large enough that $\varepsilon_f/f \leq \gamma/3$:*

(i) $\Delta\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\| \leq \nu_f$,

(ii)
$$\begin{aligned} &\mathbb{1}\big\{\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\| \geq \varphi_f\big\}\, \mathbf{E}[\Delta\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\| \mid \mathbf{Z}(t^{\mathtt{1st}}_{f,b})] \\ &\leq -\rho\, \mathbb{1}\big\{\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\| \geq \varphi_f\big\}, \end{aligned}$$

where $\rho \triangleq \gamma/3$, $\nu_f \triangleq 2\sqrt{K}f$, and $\varphi_f \triangleq 4Kf^3/\rho$.

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.

Proof. The proof, which follows similar steps as in [6], is omitted due to space limitations. □

Lemma 2. *For frames $f \geq f_0 > 0$ and blocks $b = 0, 1, \ldots, b_f$ where f_0 is some frame large enough that $\varepsilon_{f_0} \leq f_0\gamma/3$, the following bound on the moment-generating function of $\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\|$ holds:*

$$\mathbf{E}\Big[e^{\zeta_f \|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\|}\Big] \leq \mathbf{E}\Big[e^{\zeta_{f_0} \|\mathbf{Z}(t^{\mathtt{1st}}_{f_0,0})\|}\Big] + \frac{2\, e^{\zeta_f\,(\varphi_f+\nu_f)}}{\zeta_f\,\rho} \tag{19}$$

where $\rho \triangleq \gamma/3$, $\nu_f \triangleq 2\sqrt{K}f$, $\varphi_f \triangleq 4Kf^3/\rho$, and $\zeta_f \triangleq \rho\,/\left(\nu_f^2 + \nu_f\rho/3\right)$.

Proof. We use the drift conditions from Lemma 1 to prove the bound by induction using some similar steps as in [6]. When checking the inductive step, careful attention must be paid to the edge-case of transitioning to a new frame. The full proof is omitted due to space limitations. □

Theorem 1. *Under POSS, $\mathcal{V}_k(t) = o(1)$ for all $k \in [K]$.*

Proof. Fix a timeslot t large enough that its containing frame $f \geq f_0$ where f_0 is some frame large enough that $\varepsilon_{f_0} \leq f_0\gamma/3$. Then applying Jensen's inequality to our bound (19) in Lemma 2 gives

$$\begin{aligned}
&\mathbf{E}\big[\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\|\big] \\
&\leq \frac{1}{\zeta_f}\log\left(\mathbf{E}\Big[e^{\zeta_{f_0}\|\mathbf{Z}(t^{\mathtt{1st}}_{f_0,0})\|}\Big] + \frac{2\,e^{\zeta_f\,(\varphi_f+\nu_f)}}{\zeta_f\,\rho}\right) \\
&\leq \frac{1}{\zeta_f}\log\left(\mathbf{E}\Big[e^{\zeta_{f_0}\|\mathbf{Z}(t^{\mathtt{1st}}_{f_0,0})\|}\Big] + \frac{2}{\zeta_f\,\rho}\right) + \varphi_f + \nu_f \\
&\overset{(a)}{\leq} \underbrace{\left[\frac{16K}{3\rho}\log\left(\mathbf{E}\Big[e^{\zeta_{f_0}\|\mathbf{Z}(t^{\mathtt{1st}}_{f_0,0})\|}\Big] + \frac{32K}{3\rho^2}\right) + \frac{48K}{3\rho}\right]}_{\triangleq\, c} f^3
\end{aligned} \tag{20}$$

where (a) is due to several algebraic manipulations. The key here is that the f^3 term due to φ_f dominates the bound. Now, recalling Remark 2, for all $k \in [K]$, we have

$$\begin{aligned}
\mathcal{V}_k(t) &\leq \Big[\mathbf{E}[Z_k(t^{\mathtt{1st}}_{f,b})] - (f-1)^3\log^{\frac{1}{3}} f + f\Big]^+ \\
&\leq \Big[\mathbf{E}\big[\|\mathbf{Z}(t^{\mathtt{1st}}_{f,b})\|\big] - (f-1)^3\log^{\frac{1}{3}} f + f\Big]^+ \\
&\leq \Big[c\,f^3 - (f-1)^3\log^{\frac{1}{3}} f + f\Big]^+ \overset{(a)}{=} o(1)
\end{aligned} \tag{21}$$

where (a) is because $f \to \infty$ as $t \to \infty$ and therefore $\lim_{t\to\infty}\Big[c\,f^3 - (f-1)^3\log^{\frac{1}{3}} f + f\Big]^+ = 0$. □

B. Sublinear Regret

In this subsection, we show that POSS achieves $\widetilde{O}(T^{3/4})$ regret. We first bound the switching regret, using a technique that only relies on the block schedule and not on how each particular arm is chosen at the beginning of a block. Therefore, the technique is reusable for other challenging bandit with switching cost problems that are not able to use an instance-dependent analysis. This technique is also used in parts of our reward regret bound.

Theorem 2. *Under POSS, $R^{\mathtt{switch}}(T) = O\big(T^{3/4}\big)$.*

Proof. Consider the frame f such that $t^{\mathtt{fin}}_{f-1,b_{f-1}} \leq T \leq t^{\mathtt{fin}}_{f,b_f}$. In the worst case, assume we switch in every block. The total number of blocks contained in frames $1, 2, \ldots, f$ can be bounded by

$$\begin{aligned}
&\sum_{\ell=1}^{f}\left\lceil\left(\ell^4 - (\ell-1)^4\right)/\ell\right\rceil \\
&\leq \sum_{\ell=1}^{f}\left(\left(\ell^4 - (\ell-1)^4\right)/\ell + 1\right) = f + \sum_{\ell=1}^{f}\frac{\ell^4}{\ell} - \sum_{\ell=0}^{f-1}\frac{\ell^4}{\ell+1} \\
&\leq f + \sum_{\ell=1}^{f}\frac{\ell^4}{\ell} - \sum_{\ell=1}^{f-1}\frac{\ell^4}{\ell+1} = f + \frac{f^4}{f} + \sum_{\ell=1}^{f-1}\frac{(\ell+1)\ell^4 - \ell\,\ell^4}{\ell(\ell+1)} \\
&\leq f + f^3 + \sum_{\ell=1}^{f-1}\ell^2 \leq 3f^3
\end{aligned} \tag{22}$$

and there are N blocks in frame $f = 0$. Then

$$\begin{aligned}
\frac{R^{\mathtt{switch}}(T)}{T^{3/4}} &= \frac{C\sum_{t=1}^{T}\mathbf{E}[H(t)]}{T^{3/4}} \leq \frac{C\sum_{t=1}^{t^{\mathtt{fin}}_{f,b_f}}\mathbf{E}[H(t)]}{(t^{\mathtt{fin}}_{f-1,b_{f-1}})^{3/4}} \\
&\leq \frac{C(N+3f^3)}{((f-1)^4)^{3/4}} = \frac{C(N+3f^3)}{(f-1)^3} = O(1).
\end{aligned} \tag{23}$$

So, we conclude that $R^{\mathtt{switch}}(T) = O\big(T^{3/4}\big)$. □

Before analyzing the reward regret, we will need to consider another construction called an *ε-tight* Markovian policy, which is designed to incur regret on the order $\varepsilon T/\gamma$ while satisfying (2) when γ is replaced by ε.

First, we introduce the γ-tight Markovian policy $i^\gamma(t)$, which is characterized by the stationary distribution $\boldsymbol{\sigma}^\gamma$ that satisfies (2) and the stochastic matrix $\mathbf{P}^{(\boldsymbol{\sigma}^\gamma,0)}$. From here, we can derive the ε-tight policy $i^\varepsilon(t)$ for any $\varepsilon \in (0,\gamma]$ characterized by the stationary distribution $\boldsymbol{\sigma}^\varepsilon \triangleq \left(1 - \frac{\varepsilon}{\gamma}\right)\boldsymbol{\sigma}^* + \frac{\varepsilon}{\gamma}\boldsymbol{\sigma}^\gamma$ and the stochastic matrix $\mathbf{P}^{(\boldsymbol{\sigma}^\varepsilon,0)}$.

Remark 4. Observe that under the policy $i^\varepsilon(t)$, we have

$$\sum_{i\in[N]}\sigma_i^\varepsilon w_{i,k} = \left(1 - \frac{\varepsilon}{\gamma}\right)\underbrace{\sum_{i\in[N]}\sigma_i^* w_{i,k}}_{\leq 0} + \frac{\varepsilon}{\gamma}\underbrace{\sum_{i\in[N]}\sigma_i^\gamma w_{i,k}}_{\leq -\gamma} \leq -\varepsilon \tag{24}$$

for all $k \in [K]$ and that the per-timeslot reward regret of policy $i^\varepsilon(t)$ is bounded by

$$\begin{aligned}
\mathbf{E}\big[X_{i^*(t)}(t) - X_{i^\varepsilon(t)}(t)\big] &= \sum_{i\in[N]}\left[\sigma_i^* - \sigma_i^\varepsilon\right]\mu_i \\
&= \sum_{i\in[N]}\left[\sigma_i^* - \left(1 - \frac{\varepsilon}{\gamma}\right)\sigma_i^* + \frac{\varepsilon}{\gamma}\sigma_i^\gamma\right]\mu_i \\
&= \frac{\varepsilon}{\gamma}\sum_{i\in[N]}\left[\sigma_i^* - \sigma_i^\gamma\right]\mu_i \leq \frac{\varepsilon}{\gamma}.
\end{aligned} \tag{25}$$

Now we are ready to bound the reward regret of our

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.

algorithm.

Theorem 3. *Under POSS,* $R^{\texttt{reward}}(T) = \widetilde{O}\big(T^{3/4}\big)$.

Proof. Consider the first frame f' satisfying $\varepsilon_{f'} \leq \gamma f'$ and consider the frame f containing the time horizon T, i.e. $t^{\texttt{fin}}_{f-1,b_{f-1}} \leq T \leq t^{\texttt{fin}}_{f,b_f}$. If $f \leq f'$, we simply bound $R(T) \leq t^{\texttt{fin}}_{f',b_{f'}}$. Assume then that $f > f'$. Define $\overline{\varepsilon_f} \triangleq \varepsilon_f / f$ and consider the $\overline{\varepsilon_f}$-tight policy $i^{\overline{\varepsilon_f}}(t)$. We have

$$\begin{aligned} R^{\texttt{reward}}(T) \leq\ & t^{\texttt{fin}}_{f',b_{f'}} \\ & + \underbrace{\sum_{\ell=f'}^{f} \sum_{b=0}^{b_\ell} \sum_{\tau=t^{\texttt{1st}}_{\ell,b}}^{t^{\texttt{fin}}_{\ell,b}} \Big[\mathbf{E}\Big[X_{i^*(\tau)}(\tau) - X_{i^{\overline{\varepsilon_\ell}}(\tau)}(\tau)\Big]\Big]^+}_{(a)} \\ & + \underbrace{\sum_{\ell=f'}^{f} \sum_{b=0}^{b_\ell} \sum_{\tau=t^{\texttt{1st}}_{\ell,b}}^{t^{\texttt{fin}}_{\ell,b}} \Big[\mathbf{E}\Big[X_{i^{\overline{\varepsilon_\ell}}(\tau)}(\tau) - X_{i(\tau)}(\tau)\Big]\Big]^+}_{\overline{\varepsilon}\text{-tight reward regret } R^{\overline{\varepsilon}}(t^{\texttt{fin}}_{f,b_f})}. \end{aligned} \tag{26}$$

Now, due to (25) we have

$$\begin{aligned} (a) &\leq \sum_{\ell=f'}^{f} \sum_{b=0}^{b_\ell} \sum_{\tau=t^{\texttt{1st}}_{\ell,b}}^{t^{\texttt{fin}}_{\ell,b}} \frac{\varepsilon_\ell}{\gamma \ell} = \frac{1}{\gamma} \sum_{\ell=f'}^{f} \sum_{b=0}^{b_\ell} \varepsilon_\ell \\ &= \frac{1}{\gamma} \sum_{\ell=f'}^{f} \Big(\ell^3 \log^{\frac{1}{3}}(\ell+1) - (\ell-1)^3 \log^{\frac{1}{3}} \ell\Big) \\ &\leq \frac{f^3 \log^{\frac{1}{3}}(f+1)}{\gamma} \end{aligned} \tag{27}$$

and therefore

$$R^{\texttt{reward}}(T) \leq t^{\texttt{fin}}_{f',b_{f'}} + \frac{f^\beta \log^{\frac{1}{3}}(f+1)}{\gamma} + R^{\overline{\varepsilon}}(t^{\texttt{fin}}_{f,b_f}). \tag{28}$$

The $\overline{\varepsilon}$-tight reward regret $R^{\overline{\varepsilon}}(t^{\texttt{fin}}_{f,b_f})$ can be bounded by

$$\begin{aligned} R^{\overline{\varepsilon}}(t^{\texttt{fin}}_{f,b_f}) \leq\ & \sqrt{72N \log(3Nf^4)} f^{5/2} \\ & + \frac{\mathbf{E}[V(t^{\texttt{1st}}_{f',0})]}{f'^2} + (6K+2N) f^3. \end{aligned} \tag{29}$$

(derivation omitted due to space limitations). Then overall,

$$R^{\texttt{reward}}(T) \leq c \log^{\frac{1}{3}}(f+1)\, f^3 \tag{30}$$

where $c \triangleq t^{\texttt{fin}}_{f',b_{f'}} + \frac{1}{\gamma} + \sqrt{72N} + \frac{\mathbf{E}\big[V(t^{\texttt{1st}}_{f',0})\big]}{f'^2} + 6K + 2N$. So,

$$\begin{aligned} \frac{R^{\texttt{reward}}(T)}{\log^{\frac{1}{3}}(T)\, T^{3/4}} &\leq \frac{c \log^{\frac{1}{3}}(f+1)\, f^3}{\log^{\frac{1}{3}}(t^{\texttt{fin}}_{f-1,b_{f-1}}) \left(t^{\texttt{fin}}_{f-1,b_{f-1}}\right)^{3/4}} \\ &\overset{(a)}{\leq} \frac{c \log(f+1)\, f^3}{4^{\frac{1}{3}} \log^{\frac{1}{3}}(f-1)\,(f-1)^3} = O(1). \end{aligned} \tag{31}$$

where (a) is due to (8) and the above implies the result. □

VI. Simulation Results

In this section, we perform trace-based and synthetic simulations to compare the performance of our algorithm POSS against other algorithms in the literature.

A. Trace-Based Simulations

We perform trace-based simulations in the wireless network selection setting using throughput traces collected from the 5G network providing mmWave service and two different Wi-Fi access points. The 5G trace was collected by Narayanan et al. [39] in an enclosed indoor location with an average throughput of 442Mbps and we collected throughput traces from two Wi-Fi access points in enclosed indoor locations with average throughputs 257Mbps and 99Mbps respectively. Since the average round-trip time is about 50ms, we choose this as the duration of a timeslot. Since the throughput of any network never exceeds 750Mbps, we use this as the normalizing constant to derive the service probabilities $\boldsymbol{\mu}(t)$ in each timeslot t according to the trace.

Thantharate et al. [38] measured that the 5G mmWave network drains 572mAh of battery charge on average after one hour of simultaneous voice call and video streaming, while Wi-Fi only drains 416mAh. So, we set the battery consumption rates $\boldsymbol{\beta}$ accordingly (normalized to the range [0,1]). We assume the user allows a maximum battery consumption rate of 525mAh.

We compare our algorithm against two algorithms inspired by recent work. For the first, we consider a standard (not block-based) pessimistic-optimistic algorithm similar to the one presented by Liu et al. [6], which we refer to as PO. In each timeslot t, PO plays

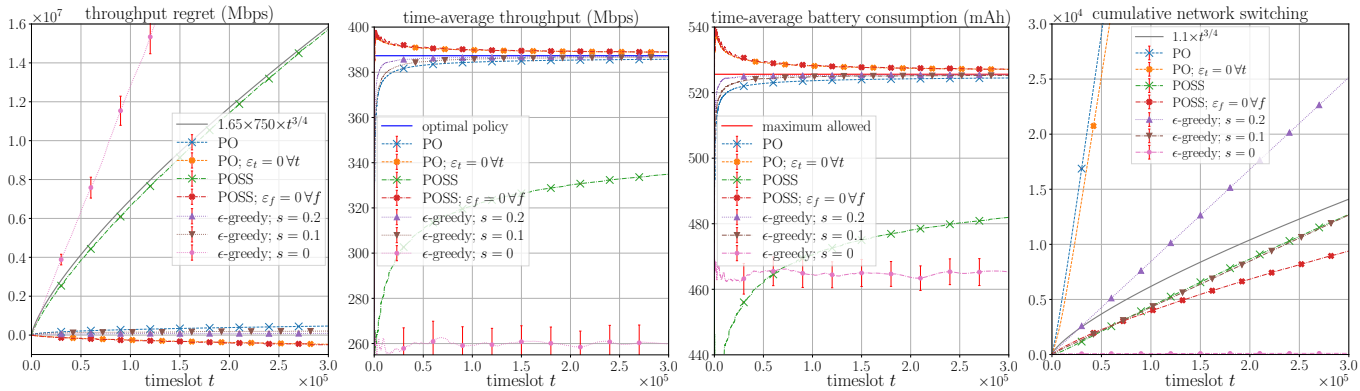
$$i(t) \in \operatorname*{argmax}_{i \in [N]} \left(\sqrt{t}\, U_i(t) - \sum_{k \in [K]} w_{i,k}\, Z_k(t) \right) \tag{32}$$

and updates its virtual queues as $Z_k(t+1) = [Z_k(t) + w_{i(t),k} + \varepsilon_t]^+$ with $\varepsilon_t = \frac{1}{\sqrt{t}}$. We also consider PO with $\varepsilon_t = 0$ for all timeslots t and POSS with $\varepsilon_f = 0$ for all frames $f > 0$ to more clearly see the impact of the block and frame mechanism.

For the second algorithm, we consider an ϵ-greedy algorithm inspired by the one presented by Krishnasamy et al. [16]. This algorithm maintains estimates $\hat{\boldsymbol{\mu}}(t)$ of the mean rewards and in each timeslot t solves the linear program (5) and plays according to the Markovian policy characterized by $\hat{\boldsymbol{\sigma}}(t)$ and $\mathbf{P}^{(\hat{\boldsymbol{\sigma}}(t),s)}$ with probability $1 - \frac{2\log t}{t}$ or explores uniformly at random with probability $\frac{2\log t}{t}$. We consider the ϵ-greedy algorithm for various values of s to see the tradeoff between reward, switching, and constraint violation. Since any $s > 0$ should incur linear regret, we also try the extreme case $s = 0$.

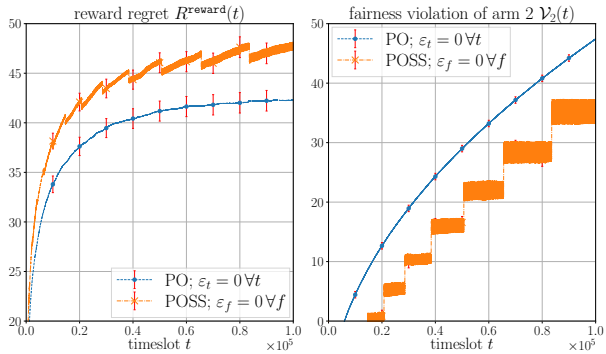
Figure 2 shows the results of our experiments, which were each run 100 times with their means plotted over time as well as their 95% confidence intervals (red error bars). Figure 2a plots the throughput regret, i.e. the cumulative per-timeslot difference in throughput between the optimal policy and each algorithm. Figures 2b and 2c plot the time-average throughput

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.



(a) Cumulative throughput regret (b) Time-average throughput (c) Time-average battery consumption (d) Cumulative switching

Fig. 2. Trace-based simulations comparing POSS to PO and ϵ-greedy in the wireless network switching setting.



(a) Reward regret (b) Constraint violation of arm 2

Fig. 3. Synthetic simulations in the bandits with fairness constraints setting.

and battery consumption respectively. Figure 2d plots the total number of switches over time.

The throughput regret of POSS appears to match the grey $O(T^{3/4})$ curve, which agrees with Theorem 3. ϵ-greedy with $s > 0$ and PO both achieve small, positive and increasing throughput regret and satisfy the battery constraint. POSS with $\varepsilon_f = 0$ and PO with $\varepsilon_t = 0$ both achieve negative throughput regret, but violate the constraint, with the constraint violation increasing sublinearly. The cumulative switching of POSS appears bounded by the grey $O(T^{3/4})$ curve, which agrees with Theorem 2, and POSS with $\varepsilon_f = 0$ achieves even lower cumulative switching. ϵ-greedy with $s > 0$ and PO both suffer linear cumulative switching. The performance of ϵ-greedy with $s = 0.1$ appears close to that of POSS here, but the curve is actually linear with a small slope. Finally, ε-greedy with $s = 0$ achieves logarithmic cumulative switching due to the evolution of the exploration probability, but suffers linear throughput regret.

B. Synthetic Simulations

To examine the impact of imposing the block mechanism on a pessimistic-optimistic algorithm, we compare POSS with $\varepsilon_f = 0$ and PO with $\varepsilon_t = 0$ in the bandits with fairness constraints setting [18]. We consider $N = 3$ arms with mean rewards $\boldsymbol{\mu} = (0.4, 0.5, 0.7)$, and fairness requirements (minimum selection fractions) $\mathbf{r} = (0, 0.3, 0.1)$. Figure V-B shows that the algorithms yield very similar performance in terms of reward regret and constraint violation when the virtual queue parameters ε are ignored. It is interesting to see how the block mechanism affects the curve. We can clearly tell from the sharp drops in regret (Figure 3a) and jumps in violation (Figure 3b) when a new frame begins. This is due to the evolution from frame-to-frame of the weight f^3 on the UCB term. We omit the figures showing the violation of the other arms as they are always zero and the figure showing the switching regret, as it is similar to the trace-based simulations with POSS dramatically outperforming PO.

A final takeaway from both our synthetic and trace-based simulations is that the POSS algorithm with $\varepsilon_f = 0$ is preferable in applications where the reward regret is more important than the constraint violation vanishing, as it is competitive with both PO and ϵ-greedy. On the other hand, if the constraint violation is required to vanish, then it is better to use ε_f as given in (11).

VII. Conclusion and Future Work

In this paper, we have studied the constrained bandits with switching costs problem in the context of wireless networking applications. To solve this problem, we developed the POSS algorithm and proved that it achieves sublinear regret and vanishing constraint violation. We corroborated our theoretical results with synthetic and trace-based simulations in one of the wireless networking applications we discussed.

For future work, we would like to consider an extension of this problem with random switching delay, where the player receives no reward and suffers positive constraint cost for a random period of time after switching. This extension is relevant to many real-world applications, where the time to switch arms is non-deterministic and depends on environmental factors. We would also like to consider the extension to random and unknown constraint coefficients revealed as bandit feedback, as discussed in Section II-B.

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] F. Li, D. Yu, H. Yang, J. Yu, H. Karl, and X. Cheng, "Multi-armed-bandit-based spectrum scheduling algorithms in wireless networks: A survey," *IEEE Wireless Communications*, vol. 27, no. 1, pp. 24–30, 2020.

[2] K. Zaheer, M. Othman, M. H. Rehmani, and T. Perumal, "A survey of decision-theoretic models for cognitive internet of things (ciot)," *IEEE Access*, vol. 6, pp. 22 489–22 512, 2018.

[3] G. Caso, O. Alay, G. C. Ferrante, L. D. Nardis, M.-G. D. Benedetto, and A. Brunstrom, "User-centric radio access technology selection: A survey of game theory models and multi-agent learning algorithms," *IEEE Access*, vol. 9, pp. 84 417–84 464, 2021.

[4] Y. Deng, X. Zhou, A. Ghosh, A. Gupta, and N. B. Shroff, "Interference constrained beam alignment for time-varying channels via kernelized bandits," 2022. [Online]. Available: https://arxiv.org/abs/2207.00908

[5] J. Steiger, B. Li, and N. Lu, "Learning from delayed semi-bandit feedback under strong fairness guarantees," in *IEEE INFOCOM 2022 - IEEE Conference on Computer Communications*, 2022, pp. 1379–1388.

[6] X. Liu, B. Li, P. Shi, and L. Ying, "An efficient pessimistic-optimistic algorithm for stochastic linear bandits with general constraints," 2021.

[7] X. Zhou and B. Ji, "On kernelized multi-armed bandits with constraints," 2022. [Online]. Available: https://arxiv.org/abs/2203.15589

[8] P. Auer, N. Cesa-Bianchi, and P. Fischer, "Finite-time analysis of the multiarmed bandit problem," *Mach. Learn.*, vol. 47, no. 2–3, p. 235–256, may 2002. [Online]. Available: https://doi.org/10.1023/A:1013689704352

[9] L. Tassiulas and A. Ephremides, "Dynamic server allocation to parallel queues with randomly varying connectivity," *IEEE Transactions on Information Theory*, vol. 39, no. 2, pp. 466–478, 1993.

[10] A. Slivkins, *Introduction to Multi-Armed Bandits*, ser. Foundations and Trends in Machine Learning Series. Now Publishers, 2019. [Online]. Available: https://books.google.com/books?id=6ViCzQEACAAJ

[11] O. Dekel, J. Ding, T. Koren, and Y. Peres, "Bandits with switching costs: T2/3 regret," in *Proceedings of the Forty-Sixth Annual ACM Symposium on Theory of Computing*, ser. STOC '14. New York, NY, USA: Association for Computing Machinery, 2014, p. 459–467. [Online]. Available: https://doi.org/10.1145/2591796.2591868

[12] R. Agrawal, M. Hedge, and D. Teneketzis, "Asymptotically efficient adaptive allocation rules for the multiarmed bandit problem with switching cost," *IEEE Transactions on Automatic Control*, vol. 33, no. 10, pp. 899–906, 1988.

[13] C. Rouyer, Y. Seldin, and N. Cesa-Bianchi, "An algorithm for stochastic and adversarial bandits with switching costs," 2021. [Online]. Available: https://arxiv.org/abs/2102.09864

[14] I. Amir, G. Azov, T. Koren, and R. Livni, "Better best of both worlds bounds for bandits with switching costs," 2022. [Online]. Available: https://arxiv.org/abs/2206.03098

[15] M. Asawa and D. Teneketzis, "Multi-armed bandits with switching penalties," *IEEE Transactions on Automatic Control*, vol. 41, no. 3, pp. 328–348, 1996.

[16] S. Krishnasamy, P. T. Akhil, A. Arapostathis, S. Shakkottai, and R. Sundaresan, "Augmenting max-weight with explicit learning for wireless scheduling with switching costs," in *IEEE INFOCOM 2017 - IEEE Conference on Computer Communications*, 2017, pp. 1–9.

[17] S. Basu and S. Shakkottai, "Switching constrained max-weight scheduling for wireless networks," in *IEEE INFOCOM 2019 - IEEE Conference on Computer Communications*, 2019, pp. 2314–2322.

[18] F. Li, J. Liu, and B. Ji, "Combinatorial sleeping bandits with fairness constraints," 2019.

[19] M. J. Neely, *Stochastic Network Optimization with Application to Communication and Queueing Systems*. Morgan and Claypool Publishers, 2010.

[20] Q. Leng, S. Wang, X. Huang, Z. Shao, and Y. Yang, "Decentralized multi-agent bandit learning for intelligent internet of things systems," in *2022 IEEE Wireless Communications and Networking Conference (WCNC)*, 2022, pp. 2118–2123.

[21] X. Gao, X. Huang, Y. Tang, Z. Shao, and Y. Yang, "History-aware online cache placement in fog-assisted iot systems: An integration of learning and control," *IEEE Internet of Things Journal*, vol. 8, no. 19, pp. 14 683–14 704, 2021.

[22] K. Guo and T. Q. S. Quek, "Dynamic computation offloading in multi-server mec systems: An online learning approach," in *GLOBECOM 2020 - 2020 IEEE Global Communications Conference*, 2020, pp. 1–6.

[23] T. Huang, W. Lin, W. Wu, L. He, K. Li, and A. Zomaya, "An efficiency-boosting client selection scheme for federated learning with fairness guarantee," *IEEE Transactions on Parallel and Distributed Systems*, p. 1–1, 2020. [Online]. Available: http://dx.doi.org/10.1109/TPDS.2020.3040887

[24] W. Xia, T. Q. S. Quek, K. Guo, W. Wen, H. H. Yang, and H. Zhu, "Multi-armed bandit-based client scheduling for federated learning," *IEEE Transactions on Wireless Communications*, vol. 19, no. 11, p. 7108–7123, Nov 2020. [Online]. Available: http://dx.doi.org/10.1109/TWC.2020.3008091

[25] Y. Zhou, C. Shen, and M. van der Schaar, "A non-stationary online learning approach to mobility management," *IEEE Transactions on Wireless Communications*, vol. 18, no. 2, pp. 1434–1446, 2019.

[26] Z. Rejiba, X. Masip-Bruin, and E. Marín-Tordera, "Towards user-centric, switching cost-aware fog node selection strategies," *Future Generation Computer Systems*, vol. 117, pp. 359–368, 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167739X20330685

[27] Z. Zhu, T. Liu, Y. Yang, and X. Luo, "Blot: Bandit learning-based offloading of tasks in fog-enabled networks," *IEEE Transactions on Parallel and Distributed Systems*, vol. 30, no. 12, pp. 2636–2649, 2019.

[28] T. Le, C. Szepesvári, and R. Zheng, "Sequential learning for multi-channel wireless network monitoring with channel switching costs," *IEEE Transactions on Signal Processing*, vol. 62, no. 22, pp. 5919–5929, 2014.

[29] H. Li, T. Li, F. Li, Y. Wu, and Y. Wang, "Cumulative participant selection with switch costs in large-scale mobile crowd sensing," in *2018 27th International Conference on Computer Communication and Networks (ICCCN)*, 2018, pp. 1–9.

[30] A. Alipour-Fanid, M. Dabaghchian, R. Arora, and K. Zeng, "Multiuser scheduling in centralized cognitive radio networks: A multi-armed bandit approach," *IEEE Transactions on Cognitive Communications and Networking*, vol. 8, no. 2, pp. 1074–1091, 2022.

[31] A. Meetoo Appavoo, S. Gilbert, and K.-L. Tan, "Shrewd selection speeds surfing: Use smart exp3!" in *2018 IEEE 38th International Conference on Distributed Computing Systems (ICDCS)*, 2018, pp. 188–199.

[32] S. Guha and K. Munagala, "Multi-armed bandits with metric switching costs," in *Automata, Languages and Programming*, S. Albers, A. Marchetti-Spaccamela, Y. Matias, S. Nikoletseas, and W. Thomas, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 496–507.

[33] B. Hajek, "Hitting-time and occupation-time bounds implied by drift analysis with applications," *Advances in Applied Probability*, vol. 14, no. 3, pp. 502–525, 1982. [Online]. Available: http://www.jstor.org/stable/1426671

[34] M. J. Neely, "Energy-aware wireless scheduling with near optimal backlog and convergence time tradeoffs," in *2015 IEEE Conference on Computer Communications (INFOCOM)*, 2015, pp. 91–99.

[35] Z. Rejiba, X. Masip-Bruin, and E. Marin-Tordera, "A user-centric mobility management scheme for high-density fog computing deployments," in *2019 28th International Conference on Computer Communication and Networks (ICCCN)*, 2019, pp. 1–8.

[36] M. Shi, X. Lin, and L. Jiao, "Power-of-2-arms for bandit learning with switching costs," *MobiHoc 2022*, 2022.

[37] Q. Wu, Z. Du, P. Yang, Y.-D. Yao, and J. Wang, "Traffic-aware online network selection in heterogeneous wireless networks," *IEEE Transactions on Vehicular Technology*, vol. 65, no. 1, pp. 381–397, 2016.

[38] A. Thantharate, C. Beard, and S. Marupaduga, "An approach to optimize device power performance towards energy efficient next generation 5g networks," in *2019 IEEE 10th Annual Ubiquitous Computing, Electronics and Mobile Communication Conference (UEMCON)*, 2019, pp. 0749–0754.

[39] A. Narayanan, E. Ramadan, J. Carpenter, Q. Liu, Y. Liu, F. Qian, and Z.-L. Zhang, "A first look at commercial 5g performance on smartphones," in *Proceedings of The Web Conference 2020*, ser. WWW '20. New York, NY, USA: Association for Computing Machinery, 2020, p. 894–905. [Online]. Available: https://doi.org/10.1145/3366423.3380169

[40] A. Eryilmaz and R. Srikant, "Asymptotically tight steady-state queue length bounds implied by drift conditions," *Queueing Systems*, vol. 72, pp. 311–359, 2012.

Authorized licensed use limited to: Queen's University. Downloaded on March 12,2026 at 00:10:36 UTC from IEEE Xplore. Restrictions apply.